Th2B-5

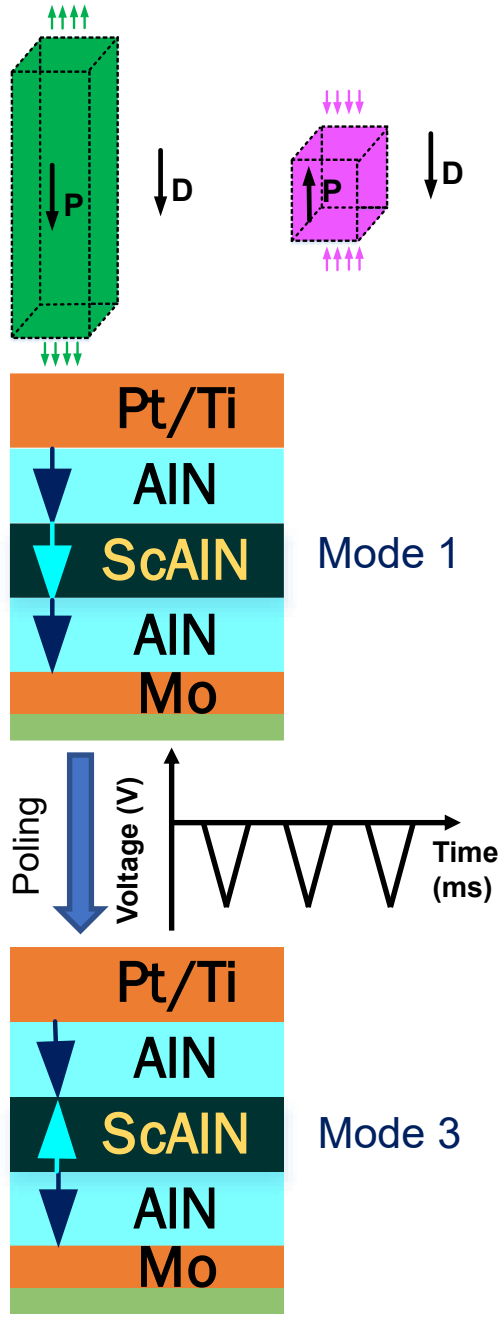
A 36 GHz Trilayer AlN/ScAlN/AlN Periodically Poled FBAR

Wenhao Peng, Suhyun Nam, Ding Wang, Zetian Mi, Amir Mortazawi

The University of Michigan, Ann Arbor, MI

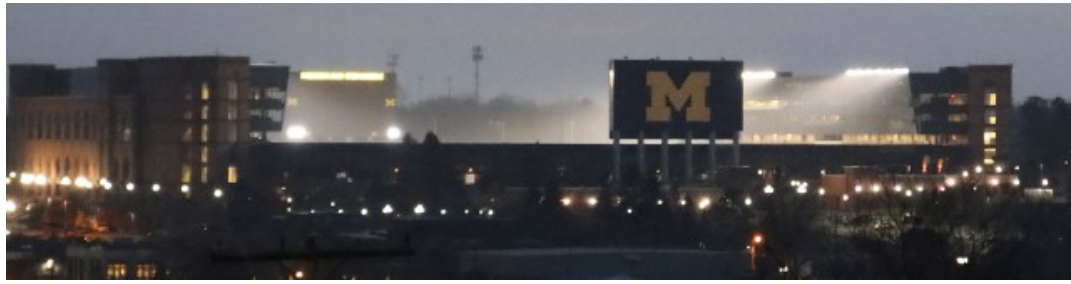
Outline

- Introduction
- Structural Analysis and Design
- Film Growth and Device Fabrication
- Polarization Switching and Impedance Response Measurements
- Conclusion

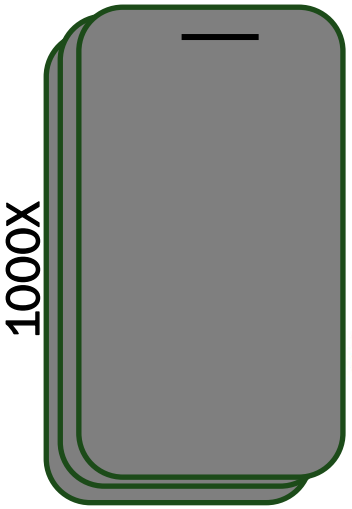


Acoustic Filters for mm-Wave Frontends

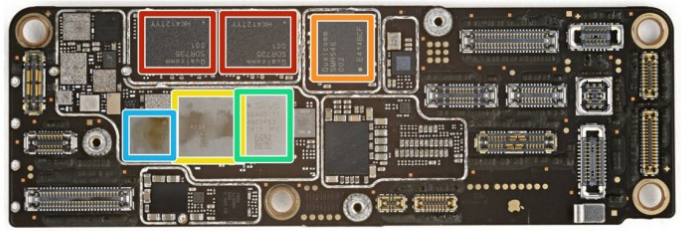
- New bands at higher frequencies require advanced filters
- Acoustic filters: technology in cellphone frontend for sub-6 bands
- Challenges for scaling conventional FBARs to mm-waves
  - Reduced film thickness and film quality
  - Increased loss and reduced coupling coefficient
- Innovations in device design and material growth enable acoustic filters at mm-wave frequencies
  - Periodically poled FBAR
  - MBE grown AlN and ScAlN



Behind the Big House stands a cellular tower



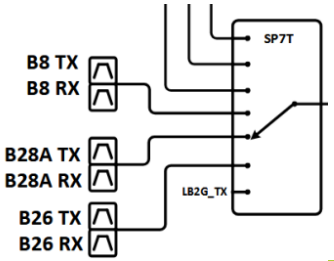
A cellular tower is accessed by thousands of smartphones



iPhone 16 Pro RF Board

- Qualcomm SDR735-001 RF transceiver
- Qualcomm SMR546-002 IF transceiver
- Broadcom AFEM-8234 front-end module
- Skyworks SKY58440-11 front-end module
- Qorvo QM76307 front-end module

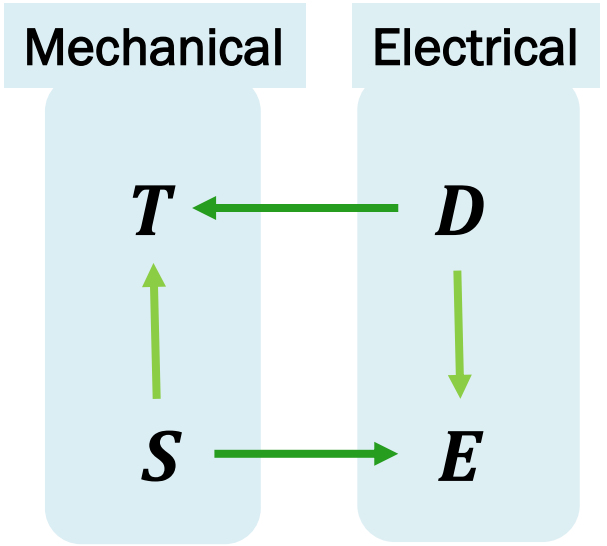
Inside one iPhone 16 Pro sits a few front-end modules



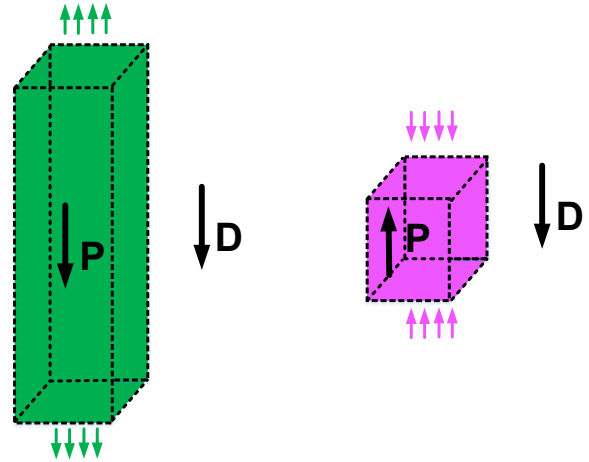
On a front-end module lies tens of acoustic filter chips

Multilayer Piezoelectric in Series

- $T = c^E S - eE$ Stress charge form
- $D = eS + \epsilon^S E$ seen in FEA
- If multiple piezoelectric layers are electrically connected in series, they share the same D
- $E = (D - eS)/\epsilon^S$
- $T = c^E S - \frac{e(D-eS)}{\epsilon^S} = c^D S - \frac{e}{\epsilon^S} D$
- "Eigen": $D = 0$
- Mode shapes are normalized and orthogonal:
  - $\langle w^{(n)} | \rho w^{(m)} \rangle = \delta_{mn}$
  - $\left\langle w_{,z}^{(n)} \middle| c^D w_{,z}^{(m)} \right\rangle = \omega_n^2 \delta_{mn}$

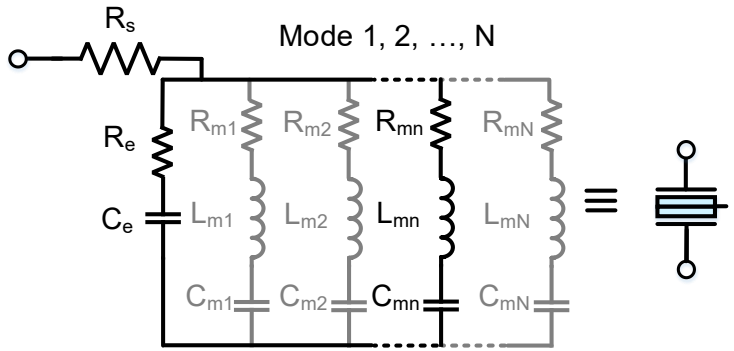


T stress E electric field
S strain D electric flux density



Tailor Piezoelectric to Higher Order Mode

- Impedance response: test source and modal analysis
- Inject $D = D_0 \cos(\omega t)$; $w = \sum_n b_n(t) w^{(n)}(z)$:
- EoM: $-\rho\omega^2 w = \left(c^D w_{,z} - eD/\epsilon^S\right)_{,z}$
- Modal amplitude $b_n(t) = D\left\langle w_{,z}^{(n)} \middle| e/\epsilon^S \right\rangle / (\omega_n^2 - \omega^2)$
- Impedance $Z = \frac{V}{I} = \int E\, dz \,/ (j\omega A D)$
- $W_n = \left[\int \left(\frac{e}{\epsilon^S}\right) w_{,z}^{(n)} dz\right]^2$ depends on material properties and mode shapes
- $k_t^2 = \frac{\pi^2}{8}\frac{\omega_p^2 - \omega_s^2}{\omega_p^2} = \frac{\pi^2}{8}\left(\frac{\left\langle w_{,z} \middle| e/\epsilon^S \right\rangle^2}{\int (1/\epsilon^S) dz \left\langle w_{,z} \middle| c^D w_{,z} \right\rangle}\right)$
- Physically k_t^2 := ratio between converted electric energy and supplied elastic energy
- Design formula for periodically poled FBAR



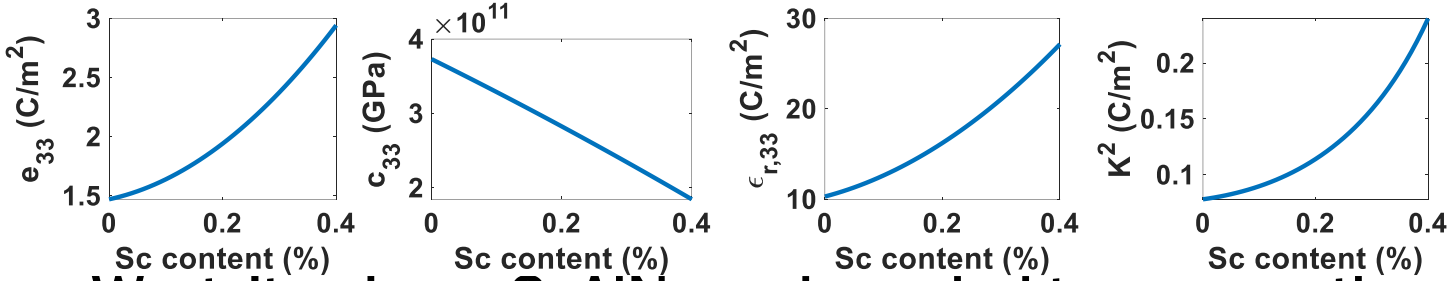
$$C_e = \frac{A}{\int \left(\frac{1}{\epsilon^S}\right) dz} \qquad L_{mn} = \frac{A}{W_n C_e^2}$$

$$C_{mn} = \frac{W_n C_e^2}{\omega_n^2 A - W_n C_e} \qquad R_{mn} = \frac{\omega_p A}{C_e^2 Q_n W_n}$$

Each mechanical branch RLC represents one mode; coupling coefficient in a higher order mode can be optimized by tailoring the sign of piezoelectric coefficients according to spatial derivative of the mode shape

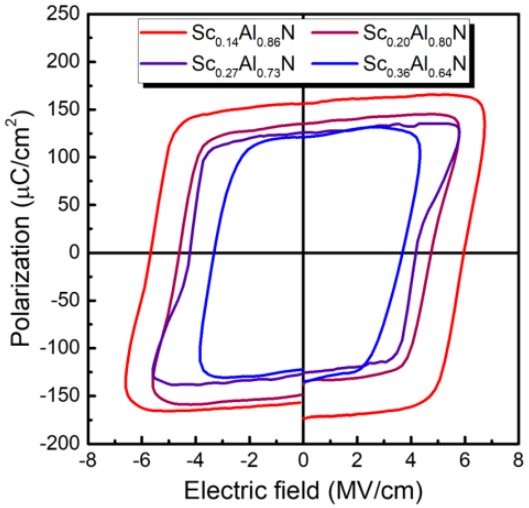
Wurtzite Phase ScAlN

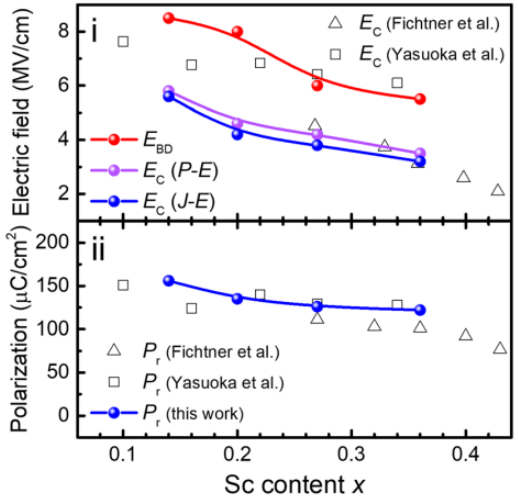
- Sc doping enhances wurtzite AlN piezoelectric properties

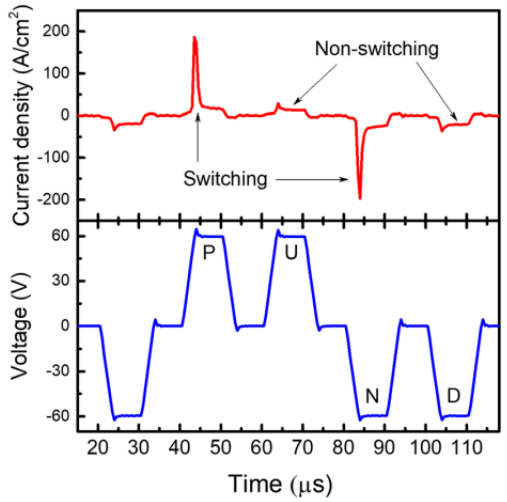


O. Ambacher, B. Christian, N. Feil, D. F. Urban, C. Els¨asser,M. Prescher, and L. Kirste, "Wurtzite ScAlN, InAlN, and GaAlN crystals, a comparison of structural, elastic, dielectric, and piezoelectric properties," Journal of Applied Physics, vol. 130, no. 4, p. 045102, 072021.

- Wurtzite phase ScAlN can be poled to reverse the sign of piezoelectric coefficient
- Remnant polarization can be measured by PUND



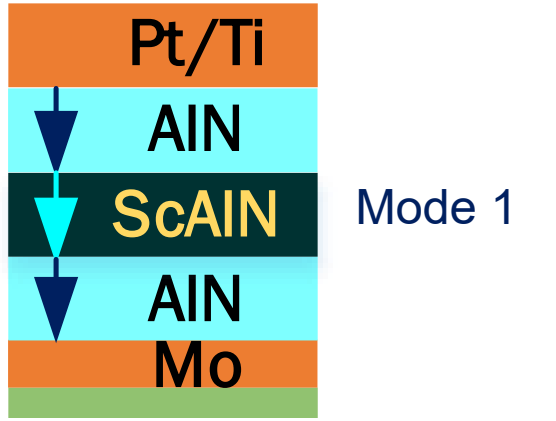




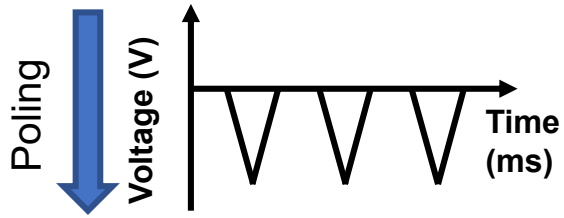
P. Wang, D. Wang, N. M. Vu, T. Chiang, J. T. Heron, and Z. Mi,"Fully epitaxial ferroelectric scaln grown by molecular beam epitaxy,"Applied Physics Letters, vol. 118, no. 22, p. 223504, 06 2021

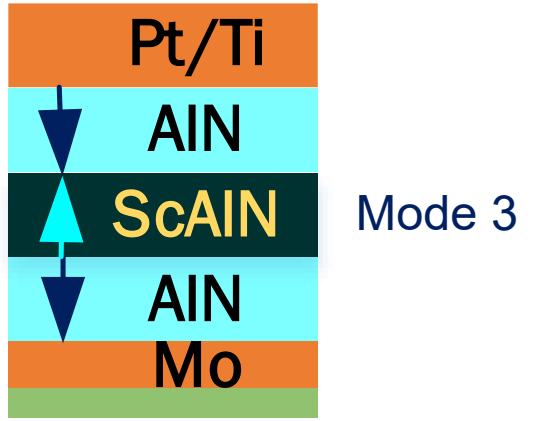
Trilayer Piezoelectric for Third Mode

- Promising approach for preserving k_t^2 and Q at mm-waves
  - Periodically Poled AlN/ScAlN/AlN
  - Maintains overall device thickness
  - Three layers work together
- Structure optimized for both k_t^2 and Q by focusing elastic strain energy to the MBE grown trilayer

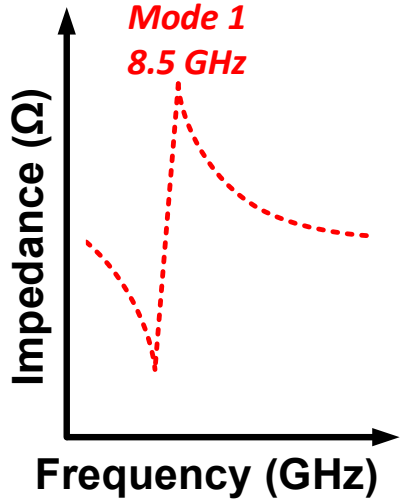


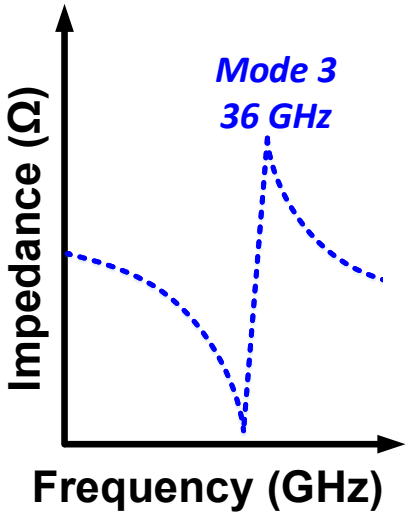
Mode 1





Mode 3



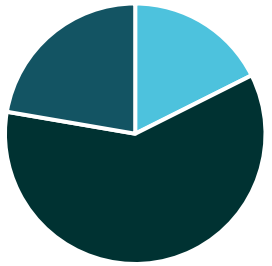


Structural Analysis and Design

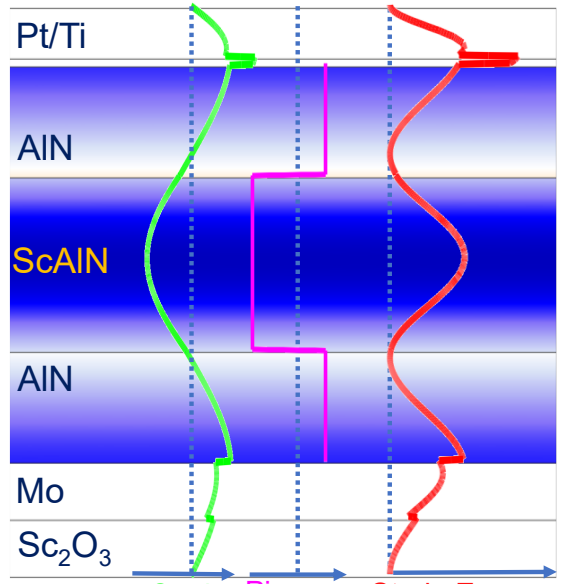
- Similar characteristic acoustic impedance between AlN and ScAlN; contrast in characteristic acoustic impedance between Mo and AlN and between Pt and AlN
- Optimize with design formula $k_t^2 = \frac{\pi^2}{8}\left(\frac{\langle w_{,z}|e/\epsilon^S\rangle^2}{\int(1/\epsilon^S)dz\langle w_{,z}|c^D w_{,z}\rangle}\right)$
- Goals: a) sign of strain generated electric field is the same inside the piezoelectric trilayer; b) Magnitude of strain energy goes down quickly outside the trilayer



$\langle w_{,z}|c^D w_{,z}\rangle$



$\langle w_{,z}|e/\epsilon^S\rangle$





Acoustic fields optimized for mode 3 at 36 GHz

Pt/Ti AlN ScAlN AlN Mo/oxide

Physical meanings of terms:

$\langle w_{,z}|c^D w_{,z}\rangle$ elastic strain energy

$\langle w_{,z}|e/\epsilon^S\rangle$ strain generated electric voltage

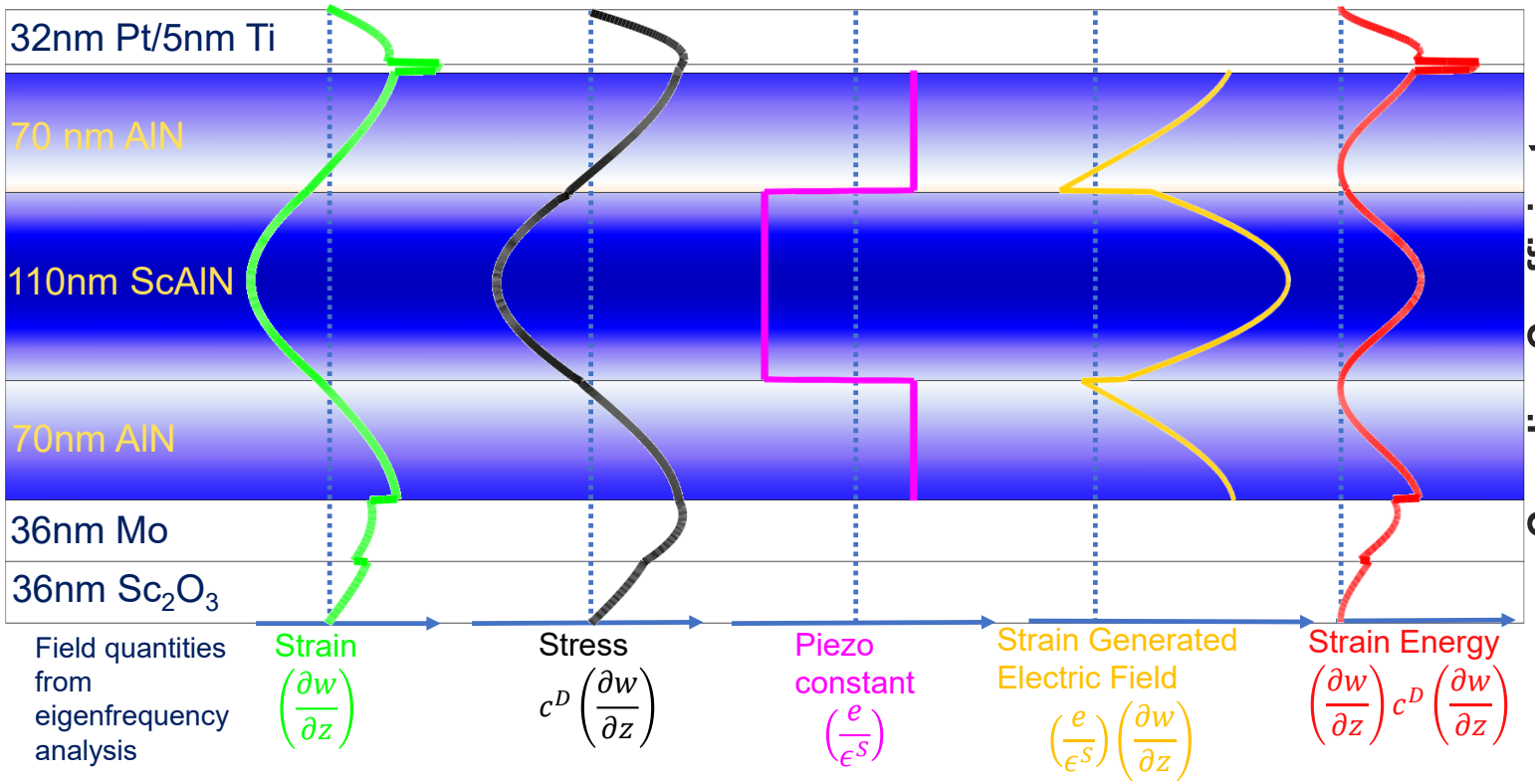
$\int(1/\epsilon^S)\,dz$ inverse of electric capacitance

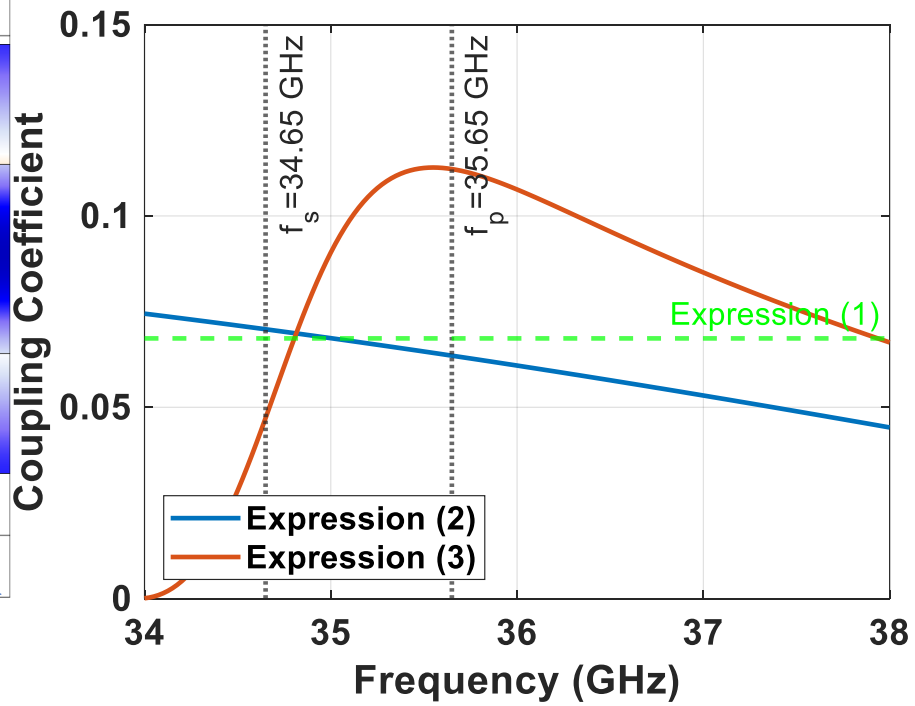
Electromechanical coupling coefficient

- $k_t^2 = \frac{\pi^2}{8}\frac{\omega_p^2-\omega_s^2}{\omega_p^2}$ (1)

$$k_t^2 = \frac{\pi^2}{8}\left(\frac{\left\langle w_{,z}\middle| e/\epsilon^S\right\rangle^2}{\int(1/\epsilon^S)dz\left\langle w_{,z}\middle| c^D w_{,z}\right\rangle}\right) \quad (2)$$

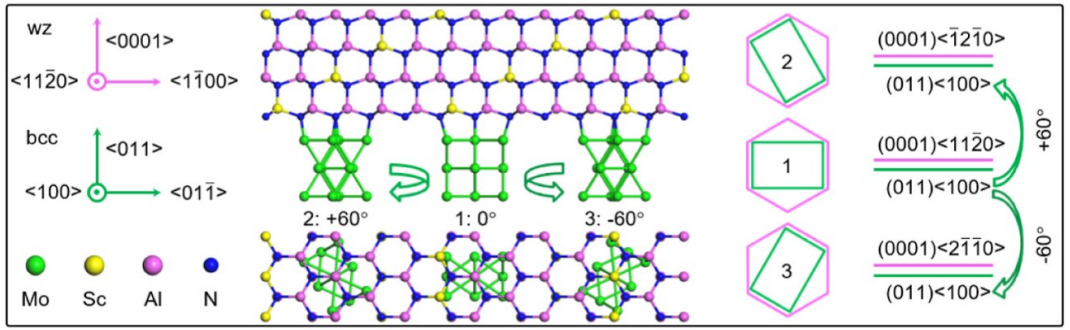
$$k_t^2 = \frac{\pi^2}{8}\frac{U_{mutual}^2}{U_E U_M} \quad (3)$$

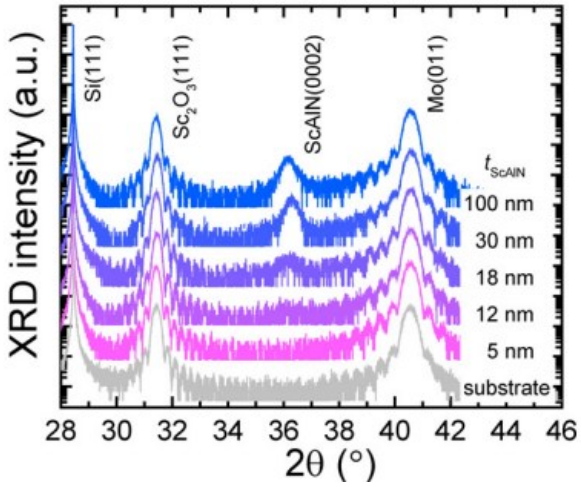


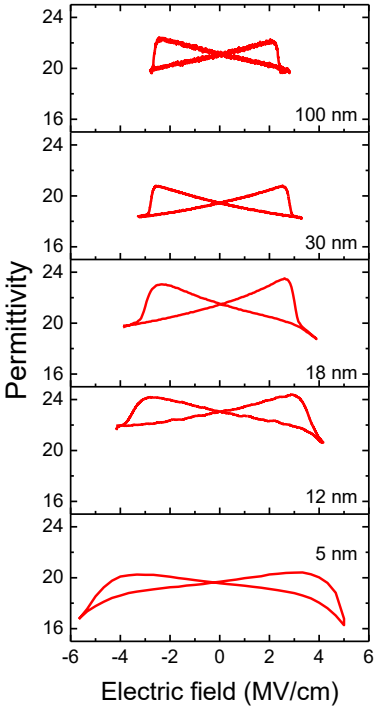


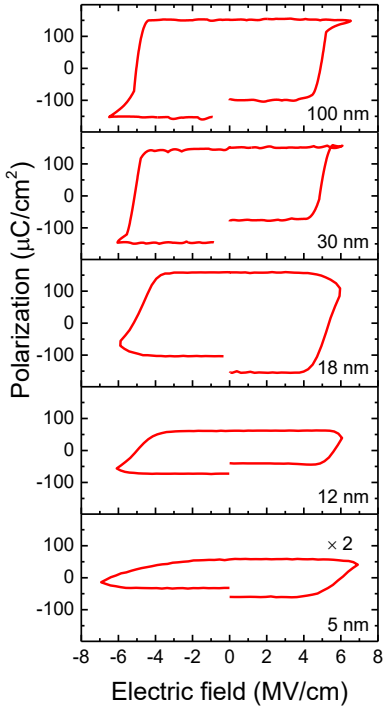
Simulation predicts parallel resonance at 36 GHz with coupling coefficient of 7% for mode 3

MBE Growth of Single Crystal Nitride









Single-crystalline wurtzite phase ScAlN was achieved on polycrystalline Mo surface

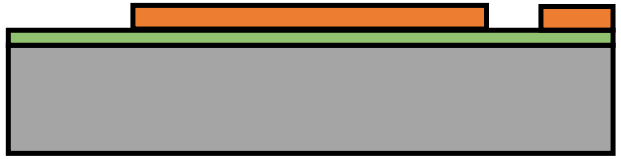
P. Wang, D. Wang, S. Mondal, M. Hu, Y. Wu, T. Ma, and Z. Mi,"Ferroelectric nitride heterostructures on cmos compatible molybdenumfor synaptic memristors," ACS Applied Materials & Interfaces, vol. 15,no. 14, pp. 18 022–18 031, 2023.

Ferroelectricity verified in single-crystalline ScAlN with high Sc contents down to 5 nm

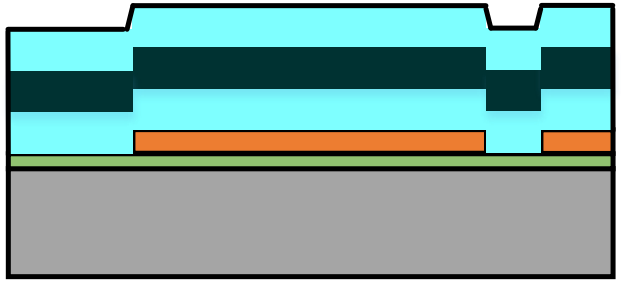
D. Wang, P. Wang, S. Mondal, M. Hu, D. Wang, Y. Wu, T. Ma, and Z. Mi, "Thickness scaling down to 5nm of ferroelectric ScAlN onCMOS compatible molybdenum grown by molecular beam epitaxy,"Applied Physics Letters, vol. 122, no. 5, p. 052101, 01 2023.

The trilayer AlN/ScAlN/AlN in this work is MBE grown using the optimized conditions developed in these works.

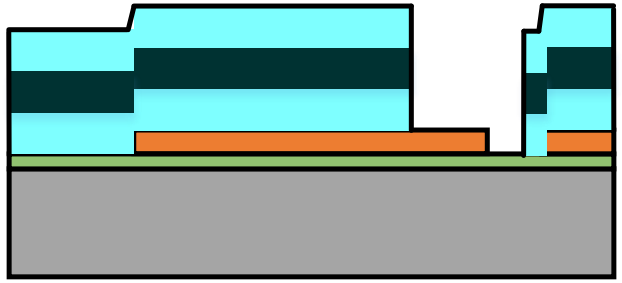
Film Growth and Device Fabrication



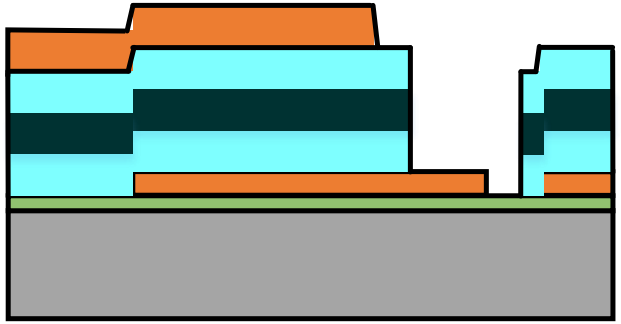
1. Dry etch patterning of Mo bottom electrode



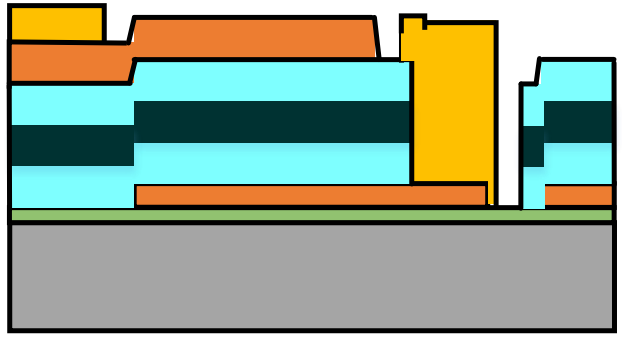
2. MBE growth of trilayer AlN/ ScAlN/AlN



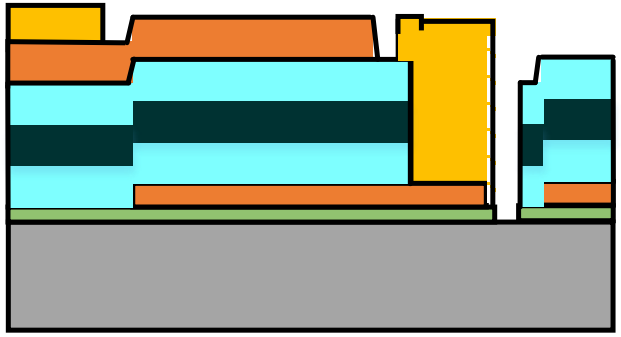
3. Plasma and wet etch of via holes in $Sc_xAl_{(1-x)}N$ layers



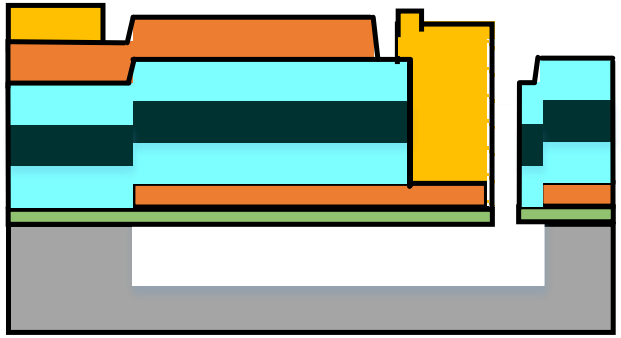
4. Physical vapor deposition of Pt electrode and liftoff



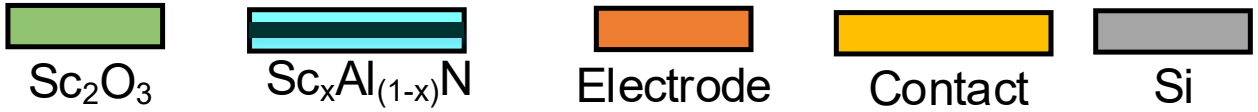
5. Physical vapor deposition of Al contact layer and liftoff



6. Plasma etch of Sc_2O_3



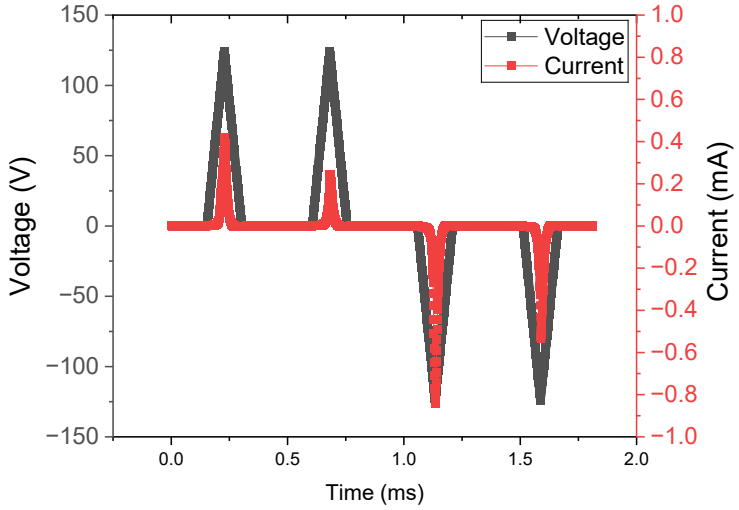
7. Dry etch of Si

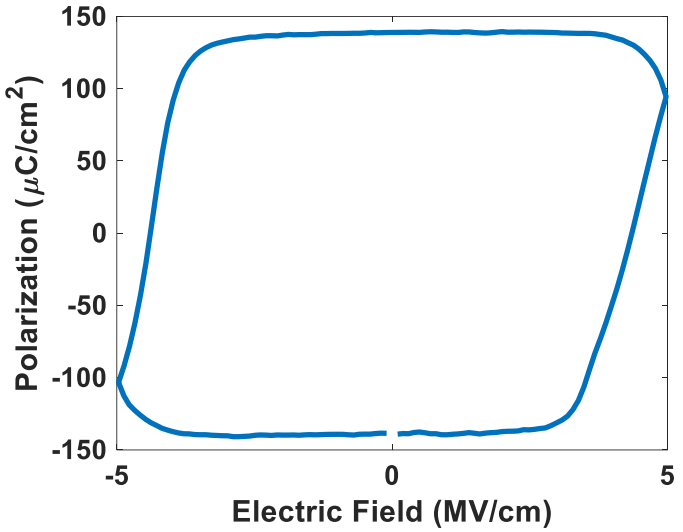


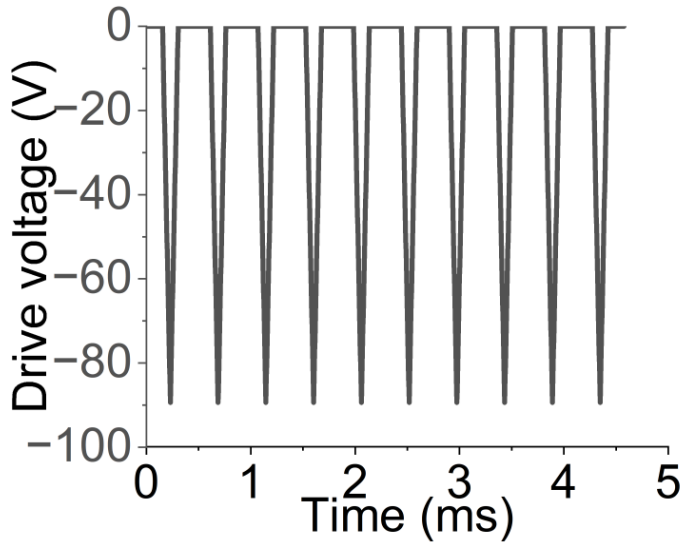
Polarization Switching

1) Remnant polarization measured by PUND: similar to single layer ScAlN measured remnant polarization

2) Choose a lower voltage pulse sequence to avoid damage but achieve the same amount of remnant polarization after switching



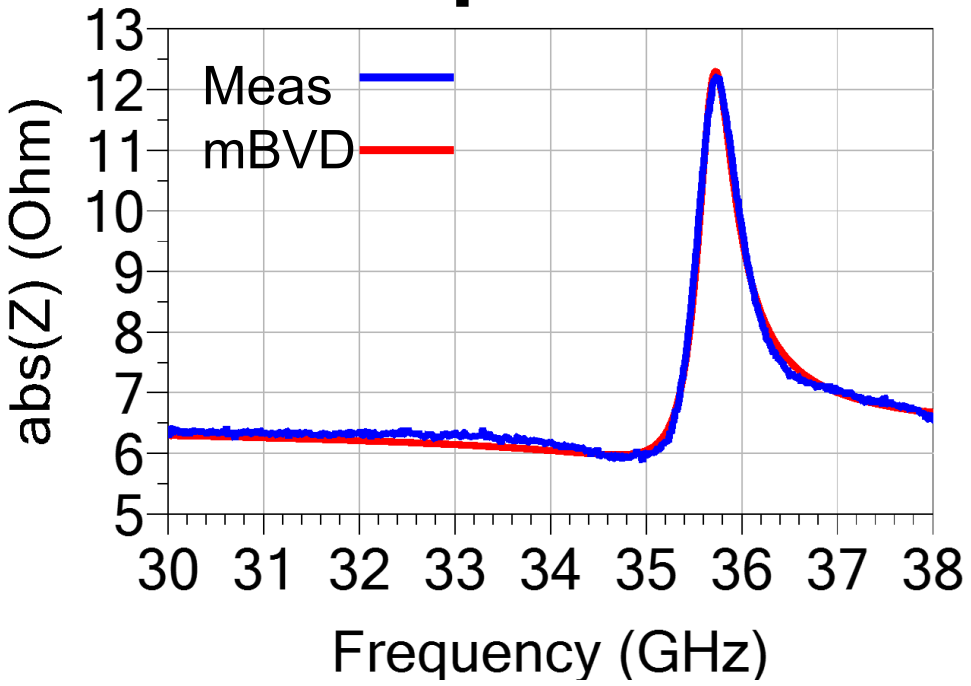


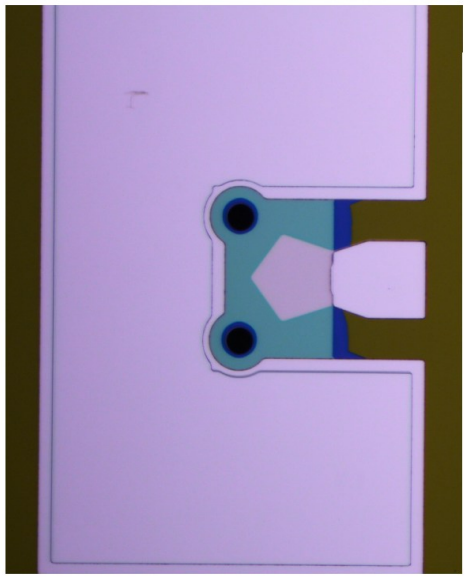


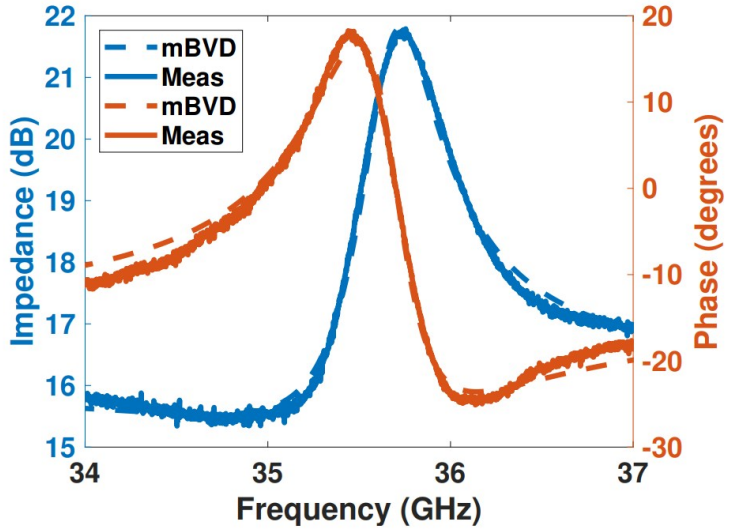
Poling waveform

3) Apply this poling waveform to FBARs and gauge impedance response: from fundamental mode at 8.5 GHz to 3rd mode at 36 GHz

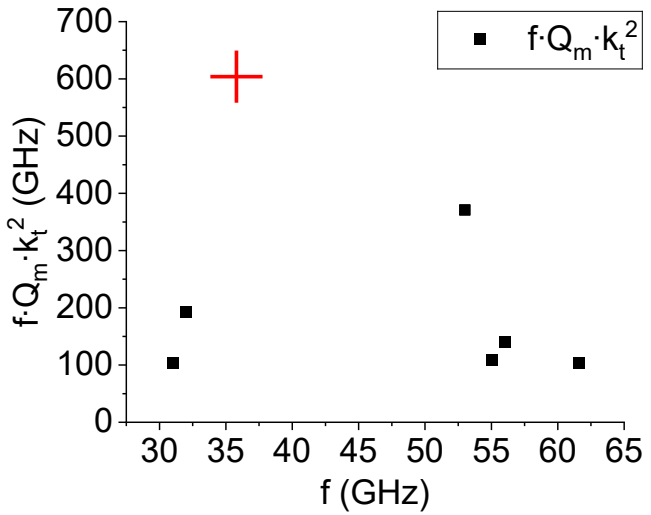
Impedance Response Measurements







Fully poled state: resonance at 35.8 GHz with $k_t^2 = 6.7\%$ and $Q_m = 252$

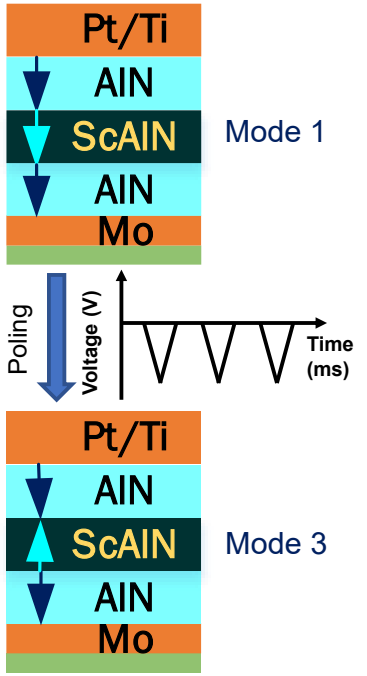


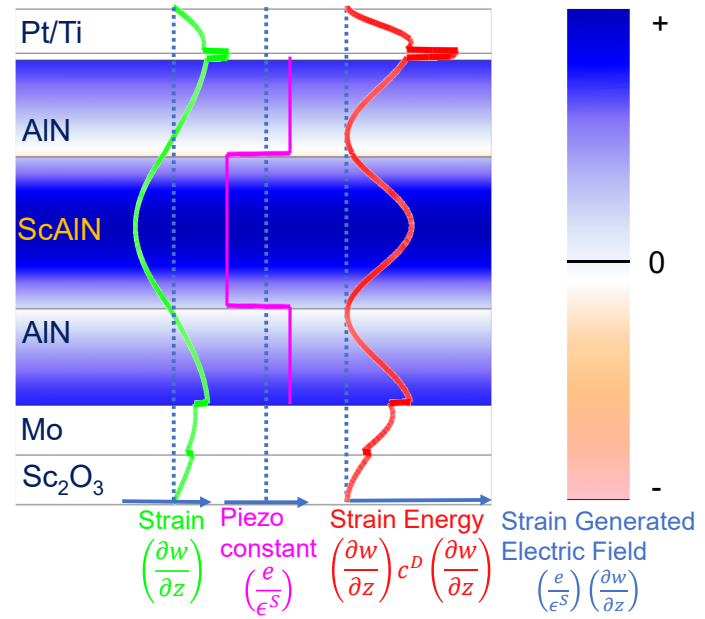
This work: best reported performance in FBARs at mm-wave

f (GHz)	Q_m	k_t^2 (%)	$Q_m \cdot k_t^2$	$f \cdot Q_m \cdot k_t^2$
35.8	***252***	***6.7***	***16.9***	***604***
53	114	6.1	7.0	371
56	60	4.2	2.5	141
55	90	2.2	2.0	109
31	61	5.5	3.4	104
61.6	94	1.8	1.7	104

Conclusion

- A new trilayer AlN/ScAlN/AlN periodically poled FBAR
- Successfully demonstrated poling and 36 GHz measurements for the third resonance mode, exhibiting best reported performance
- Future work on demonstrating mm-wave filters based on such resonators





Acoustic fields optimized for mode 3 at 36 GHz

